

INSTITUTO SUPERIOR TÉCNICO

Departamento de Engenharia Informática

Forensics Cyber-Security

MEIC, METI

Digital Forensics Report

Lab Assignment III – Operation DeepFISH – Stage III

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1 Acquired artifacts

This section lists the evidence artifacts recovered from the network captures. Only items relevant to the case questions are listed; inconclusive or background material is excluded. All times are **UTC** for `t1`/`t3` and **WEST** (UTC+1) for `t2`, taken from each capture's own timestamps and stated in its native zone (Portugal is UTC+0 until 29 Mar 2026 and UTC+1 after that; the offset only matters when comparing `t2` against the other two captures). An account or host is never equated with a person: we write "host .15 (account `tiago.estagiario`)" and defer attribution to Section 3.6.

All files below were carved from unencrypted traffic (intranet HTTP, TFTP, plaintext FTP) or from TLS sessions decrypted with the respective TLS key-log, then preserved read-only and SHA-256 hashed. The exact Wireshark display filter and the exported artifact for each item are consolidated in the tables of Section 5.1.

File	Source / provenance	SHA-256
`amelia.pdf`	`t1.pcapng`, intranet `/docs/amelia.pdf`, host .15, 3 Apr 20:15:06	`04be0e12533ede61c718f63b3bedf9888b2914dfe0e437d9c0f1272b24f51707`
`amelia_tools_config.json`	`t1.pcapng`, intranet `/docs/amelia_..._config.json` (loaded by `integracao.js`), host .15	`4cf80e2f2a1321b13e3dece45d0c949de73e5aadc3ca9d1d8e4339b398101110`
`sigma04.conf`	`t1.pcapng`, TFTP `profiles/sigma04.conf` → .15, 20:25:00	`c8e3e78522de33afe41dcee1655b35ed7db78c278e8aa1f7da677319de388713`
`sigma04_bootstrap.env`	`t1.pcapng`, TFTP `/pxe/sigma04_bootstrap.env` → .15, 20:25:00	`b27bcd4b4a9623275cd36e500c49f55038ca36651f41940f60364d47e2e16708`
`evaluation_scope.txt`	`t1.pcapng`, intranet `/docs/evaluation_scope.txt`, host .15, 20:05	`04acef3c6001b6d57883d78fcd67b379f2103d752c43ae735b0bdc829b40c383`
`evaluation_cases.csv`	`t1.pcapng`, intranet `/docs/evaluation_cases.csv`, host .15, 20:05:00 (frame 49678; `ETag` = SHA-256)	`aee43f9ba079907413d9658439a2d52ab57e90701f2af0d00d153d1cd295439f`
`/api/integrations/amelia` response	`t1.pcapng`, intranet, host .15, 20:15:17	`fe8c1de78151c662baa31c79e4d08d6fc1e5ef5733b2882937fdb7e2152592c5`

Table 1: Evidence artifacts recovered from `t1.pcapng` (intranet HTTP and unauthenticated TFTP), with in-capture provenance and SHA-256.

File	Source / provenance	SHA-256
`storage_maintenance.tar.gz.gpg`	`t2.pcapng`, FTP `RETR` from .40 → .15, 11 Apr 22:45 (also DM'd by Armando to Rui, 21:34)	`780161323b3d67c4afa5d034615c44b7cc4ab1de15de34a991e1d9302eab91f4`
`storage_maintenance.tar.gz`	`t2.pcapng`, decrypted from the `.gpg` above	`f81f209f13067d2168c5ae63a042461bc8b4b9cf273fd07e1bb8db38f260d546`

File	Source / provenance	SHA-256
	with `rfisco-maintenance.gpg`	
`ama_api_tokens.json`	`t2.pcapng`, extracted from the decrypted `storage_maintenance.tar.gz` (Google admin service-account key, `armando-sacos-admin@…`)	`9e6333a767e936973af82e68ea61023388ce6b5d8697b8652985e186a305f78f`
`chromium-cache.tar.gz`	`t2.pcapng`, FTP `RETR` from .40 → .15, 11 Apr 22:45 (Armando's cached Mattermost session)	`e99400136d7a9a87da65092d0475f383b027528389d69d2c5b1c5db093216d27`
`rfisco-maintenance.gpg`	`t2.pcapng`, FTP `RETR` from .40 → .15, 11 Apr 22:45 (Rui's GPG private key)	`0b7745d9b007eb0c6f3eb435ab1803c86c912a1d57b44a061cf01e7756e542b7`
`telegram_chat.webp`	`t2.pcapng`, `:8443` workstation-agent push .10 → .15, 11 Apr 23:30 (404 394 B; planted framing artifact)	`5fb499e82ca618def24640a1e6298512646636fae243d928f10ebdbd29e50e10`
`amelia-core.pack`	`t2.pcapng`, git clone of `git.ama.gov.pt/dti-ia/amelia-core.git` by .15, 12 Apr 23:43 (one new commit, 3e2c5c8)	`5a13185d04c1cfee11e0cc4c3d66c04df81653e2e85ba291a5fc18aa055deb64`
`rag-isotropy-release.zip`	`t2.pcapng`, uploaded to ChatGPT by .15, 13 Apr 13:35 (from Vítor's desktop, authored by Rui)	`e73208149c580627295440fd9aa1e4e8bd30a10a392621b0cb63eb39d359704a`
`Technical_Assurance.pdf`	`t2.pcapng`, intranet `/docs`, host .15, 2 May 14:50	`a6353ed37c7db210a4bd29e41fa8cc773a1a6eb4dca2805a958fd956dee7935a`
`armando-voice-message.mp3`	`t2.pcapng`, Mattermost DM .10 → .40, 11 Apr 23:24 (24.5 s;	`b953c022b612f4f07bc1fc0cbf918e06318a4268bf76902a0a69f2a2c349b29d`

File	Source / provenance	SHA-256
	transcript in Section 5.2)	
`socraques-armando-mail.eml`	`t2.pcapng`, Roundcube webmail, host .10, 30 Apr 14:00 (Minister's e-mail; see Section 5.2)	`b1ad86cbe256509bf27faaabb76e638b05e7a95d4a02c4e39105963ba0a207e2`

Table 2: Evidence artifacts recovered from `t2.pcapng` (plaintext FTP, workstation-agent push, Git smart-HTTP, proxied HTTPS and intranet HTTP), with in-capture provenance.

File	Source / provenance	SHA-256
`DuckyDonald-voice-msg.ogg`	`t3.pcapng`, Discord attachment upload (HTTP/2 `PUT` to the signed storage URL) by host .15, 18 Aug 14:11:09 (frames 15646-15687); sent to Johnny at 14:12:04 (frame 16105)	`2a434fac6c51b94a895fd6c32773290c586ff39d8a67eccc19edb2bfd4fbe985`
`7nk4q8m2rvjg1.png`	`t3.pcapng`, GET `i.redd.it/7nk4q8m2rvjg1.png` by host .15, 20 Aug 10:18:27 (frame 38304; 1989497 B)	`76df940819110fc996599a9df24d54c3c55eaab85055cf1306acc2c375ee396b`
`phishing_email.eml`	`t3.pcapng`, SMTP delivery from external IP 185.77.23.20 to `ama_mail` for `tiago.estagiario@ama.gov.pt`, 22 Aug 07:00:00 (frames 44712-44739, message in frame 44735); opened by host .15 at 07:00:04 (frame 44759); phishing e-mail linking to `fenix.dpss.inesc-id.pt` (see Section 5.2)	`9e1960dc4267198c0f60a46598d78d2a6bbfadca750e4abd67f58ad5e9ae41b1`

Table 3: Evidence artifacts recovered from `t3.pcapng` (Discord attachment upload and proxied HTTPS download), with in-capture provenance and SHA-256.

2 Report of all findings

This section summarises what the network captures contain before the per-question analysis in Section 3.

2.1 Capture, method and reliability

`t1.pcapng` spans 2026-03-05 19:33:03 to 2026-04-05 10:01:00 UTC (≈30 days 13 h), with activity concentrated in four clusters: the evening of 5 Mar, the evening of 3 Apr, the morning of 4 Apr and 5 Apr at 09:00. TLS was decrypted with the provided key-log (`tshark -o tls.keylog_file:`); the

AMÉLIA node `sigma04` (10.10.20.50) serves TLS on TCP **8080** and additionally required **Decode As 8080 → TLS** (`-d tcp.port==8080,tls`) before its HTTP/JSON became visible.

`t2.pcapng` spans 2026-04-11 21:33:43 to 2026-05-02 15:11:02 **WEST** (UTC+1). All HTTPS was decrypted with `t2-tls.keys`. Internet-bound HTTPS traverses the proxy 10.10.14.1:3128 (`egress.ama.gov.pt`) under TLS inspection (the "AMA Enterprise TLS Inspection CA"), so proxied external sessions (ChatGPT, YouTube) are readable in clear; internal TLS uses the same inspection CA. Unlike `t1`/`t3` (UTC), `t2` timestamps are native **WEST**, which we keep as-is and flag wherever the two must be compared.

`t3.pcapng` spans 2026-07-28 22:50:03 to 2026-08-25 17:31:33 UTC (≈27 days 18 h, 67 515 packets, 10 capture interfaces), with activity concentrated in the late evening of 28 Jul and early morning of 29 Jul, 18 Aug, 20 Aug, the morning of 22 Aug and 24-25 Aug. `capinfos` prints local time (UTC+1); all times here are UTC, as in `t1`. TLS was decrypted with `t3-tls.keys` (`-o tls.keylog_file:t3-tls.keys`). Internet-bound HTTPS traverses the proxy 10.10.14.1:3128 under TLS inspection, so proxied external sessions (Discord, Reddit, YouTube) are readable in clear.

The trace was split into 57 windows at every gap above 300 s between consecutive frames (`time_windows`), and each window was triaged with its IP conversations and the hosts requested over HTTP/1 and HTTP/2 (`window_triage`):

```
tshark -r t3.pcapng -E separator='|' -T fields -e frame.number -e frame.time_epoch
-e frame.time_utc \
 | awk -F'|' -v g=300 '
   NR==1 {s=$1; st=$3; p=$2; pf=$1; pl=$3; n=1; next}
   { if ($2-p > g) { print s "-" pf "\t" st "\t" pl "\t" n; s=$1; st=$3; n=0 }
p=$2; pf=$1; pl=$3; n++ }
   END { print s "-" pf "\t" st "\t" pl "\t" n }' > time_windows

cut -f1 time_windows | while read r; do
  A=${r%-*}; B=${r#*-}; W="frame.number>=$A && frame.number<=$B"
  echo "=== $r"
  tshark -r t3.pcapng -q -z "conv,ip,$W" | grep '<->'
  tshark -r t3.pcapng -o tls.keylog_file:t3-tls.keys -Y "($W) && (http.request ||
http2.headers.method)" \
    -T fields -e ip.src -e http.host -e http2.headers.authority | sort | uniq -c |
sort -r
done > window_triage
```

The Discord gateway and Mattermost WebSockets were decoded with Python scripts run on the decrypted frame payloads exported with `tshark -T fields` (zlib-stream with one shared decompressor per connection for Discord; each `posted` event matched to its `POST /api/v4/posts` request for Mattermost). Tokens and passwords are redacted in all report copies; the raw extracts are kept hashed.

2.2 Hosts and accounts

The captures involve the following AMA infrastructure and workstations. Account-to-host bindings were taken from each host's Mattermost/webmail login bodies (in `t2`, from each host's Mattermost `login_id`); the `t1` server-side user list (`GET /api/v4/users`, frame 56497) confirms the full `t1` account set.

Host / IP	Role and name
10.10.14.1	HTTP `CONNECT` proxy :3128 and DNS resolver (`/proxy.pac`, DNS answers); in t2 also `egress.ama.gov.pt`, TLS-inspecting.
10.10.20.1	Datacenter gateway and **TFTP provisioner** (serves `profiles/…` and `/pxe/…`).
10.10.20.10	`chat.ama.gov.pt` (Mattermost 10.5), `intranet.ama.gov.pt` and (t2) `git.ama.gov.pt`.
10.10.20.25	`mail.ama.gov.pt` (Roundcube webmail + SMTP).

Host / IP	Role and name
10.10.20.50	`sigma04.ama.gov.pt` - AMÉLIA node “SIGMA-NODE-04”, HTTPS :8080.
10.10.14.15	account `tiago.estagiario` (Tiago Sintaxe); host `tiago-ws`, runs “AMA-Workstation-Agent/1.0.4” on :8443 (t2).
10.10.14.22	account `ines.paralelo` (Inês Paralelo);
10.10.14.30	account `ricardo.vale` (Ricardo Vale);
10.10.14.10	account `armando.sacos` (Armando Sacos, Technical Director, the only Mattermost `system_admin`);
10.10.14.40	account `rfisco` (Rui Fisco, Senior Data Engineer); runs vsFTPd 3.0.5 (t2); periodic `/health` checks on `sigma04`.
10.10.14.45	account `vitor.algoritmo` (Vítor Algoritmo, Cloud Operations Lead); new host in t2.
Discord	account `duckydonald` (“DuckyDonald”, id 262835026388028468; e-mail duckydonald@proton.me), seen only from host .15; contact `johnny` (“Johnny”, id 218482935632332960), whose host is not in the capture.

Table 4: Hosts and the accounts observed across the captures. An account/host is not by itself a person (Section 3.6).

The Mattermost user list resolves the apparent “7 users vs 6 staff” discrepancy: the seventh account is `system-bot` (`is_bot:true`), the built-in notification bot, so there is no unidentified human user. Every account was provisioned on 2026-03-01 00:00 UTC with `create_at == last_password_update`, i.e. default passwords never changed; chat and webmail logins carry these passwords in clear inside TLS, following a predictable `<prefix>_<username>` pattern. We therefore do **not** reproduce any password in this report (relevant to Section 3.3).

2.3 Overview of activity

- **Document and configuration retrieval from host .15 (t1).** On 3 Apr, account `tiago.estagiario` downloads the evaluation files (20:05), scans 10.10.20.10/.50 (20:15), walks the intranet, and then pulls the `sigma04` provisioning files over unauthenticated TFTP (20:25). Two of these downloads (`amelia.pdf`, `amelia_tools_config.json`) are byte-identical to files hidden in earlier stages (Section 3.1).
- **AMÉLIA evaluation and the staging bypass (t1).** Across 3 Apr, host .15 interacts with the AMÉLIA procurement assistant on `sigma04:8080`, both through the web page and via scripted `curl` calls, culminating in demonstrations that a **staging bypass token** turns a NON_COMPLIANT bidder into COMPLIANT and suppresses the governance alert (Section 3.4).
- **Credential theft and document exfiltration (t2).** On 11 Apr, host .15 researches an FTP password-testing tool, brute-forces Rui’s FTP server (.40) and exfiltrates three files; the recovered set self-escalates from a single weak FTP password to Rui’s GPG private key, an admin-level Google service-account key and a cached Mattermost session (Section 3.3).
- **In-production RAG manipulation (t2).** A commit by Rui to `amelia-core.git` adds a query-translation hook that deserialises externally-supplied tensors; a toolkit authored by Rui and deployed by Vítor biases AMÉLIA toward chosen procurement vendors, which is the concrete mechanism behind EV-031 = COMPLIANT seen in t1 (Section 3.4).
- **Planted framing and political pressure (t2).** A staged “Boris”/Telegram screenshot is pushed onto Tiago’s workstation from Armando’s host; the IST technical-assurance panel’s advisory majority against acceptance is overridden by the Chair alone, under coded pressure from a Minister’s e-mail to Armando (Section 5.2).
- **Chat and e-mail.** Two t1 Mattermost exchanges (Tiago/Inês; Ricardo/Inês) and a two-message EV-031 e-mail thread between Tiago and Armando, plus the t2 exchanges and the Minister’s e-mail (Section 5.2). The Ricardo/Inês messages remain the only overtly suspicious chat content, but those hosts never touch `sigma04`, TFTP, FTP or mail in either capture.

- **Discord conversation between DuckyDonald and Johnny (t3).** From 28 Jul, host .15 is logged in to Discord as `DuckyDonald` and talks to a single contact, Johnny, in a DM channel. Johnny says he has just formatted his laptop, DuckyDonald says he is tired of government work and that it began when Johnny "wanted to do illegal stuff", and on 29 Jul Johnny writes that he knows DuckyDonald is interning at AMA. On 18 Aug host .15 uploads a voice message (`DuckyDonald-voice-msg.ogg`) and sends it to Johnny, says he is reorganising files to send to "someone asap" and that a Telegram image in Russian made him stay low, and later tells Johnny to keep the files hidden inside personal files and that a USB will be left at Alameda Metro. On 25 Aug Johnny says he collected the USB and hid the files, then discovers that his personal folder is a git repository that pushes automatically, and DuckyDonald tells him to give the USB to the police if he stops answering, naming "Tiago Sintaxe" (Section 5.2, Section 3.1). All of the Johnny/DuckyDonald conversation is on Discord; the capture shows no other channel between them.
- **Ricardo and Inês (t3).** On 24 Aug Ricardo (.30) tells Inês (.22) over the AMA Mattermost that DSI and Judicial Police staff are on the floor with imaging equipment. Inês answers that they asked for her workstation and service logs, tells him to "act normal" and to leave "the booking files" alone until six, and ends by telling him to meet at the train station and not to say another word on that terminal. On 25 Aug Inês sends Ricardo a long message denying that they touched AMÉLIA or committed a crime, accusing Armando of dealings over government tenders, and asking to cancel plans to leave (flights, money abroad). These messages continue the suspicious exchange seen in t1 and t2 (Section 5.2);
- **Phishing e-mail to host .15 (t3).** On 22 Aug, shortly after host .15 logs in to the AMA webmail, an e-mail from external IP 185.77.23.20 is delivered for `tiago.estagiario@ama.gov.pt`, presented as coming from the IST IT services and linking to a look-alike Fénix login page on `fenix.dpss.inesc-id.pt`. Host .15 opens the e-mail, loads the page 12 s later and submits the login form 13 s after that; the server answers with a redirect to the real Fénix site. The capture shows the credentials being sent, not who typed them or whether they were valid (Section 5.2).
- **Background and other activity (t3).** Host .15 downloads YouTube audio at the start of the capture and an image from Reddit on 20 Aug (Section 3.1); host .10 browses two external sites with read-only requests; firmware-update checks from all hosts are blocked by the proxy throughout and are not individually listed.

3 Analysis of relevant findings

3.1 Recovery and reconstruction of the hidden files

Which of the eight hidden files identified in the previous assignments can be recovered or reconstructed from the provided artifacts, and how? Where recovery is partial or impossible, what evidence of each file remains available, and what prevents its complete recovery?

From `t1.pcapng`, **two of the eight** hidden files identified in Stages I and II are recovered byte-for-byte, both as ordinary intranet downloads rather than as concealed payloads:

- `amelia.pdf` → intranet `/docs/amelia.pdf`, downloaded by host .15 on 3 Apr 2026 at 20:15:06. (see Section 1)
- `amelia_tools_config.json` → intranet `/docs/amelia_tools_config.json` (loaded by `integracao.js`). (see Section 1)

Both digests are **byte-identical** to the copies hidden inside João Musk's `stash.zip` in Stage II (the carriers `myzip.zip` and `polish_cow.webp` respectively), which were in turn first recovered from the phishing site in Stage I. This was verified by direct hash comparison against the Stage II manifest. We infer, that because these two files are served openly from the AMA intranet `/docs/` area ("Área geral - sem sessão pessoal"), anyone on the office network could download them without authentication. The hidden copies therefore **originate from AMA's internal document repository**.

From `t2.pcapng`, a further **four** of the eight hidden files are recovered, raising the running total to **six of eight**. Each now carries a recorded SHA-256 (Section 1), byte-identical to its Stage II copy except for `Technical_Assurance.pdf`:

- `ama_api_tokens.json` → obtained by decrypting the exfiltrated `storage_maintenance.tar.gz.gpg` with Rui's `rfisco-maintenance.gpg` (Section 3.3); it is the Google admin service-account key for `armando-sacos-admin@…`.
- `train_anisotropic_override.py` → carried inside `rag-isotropy-release.zip`, uploaded by host .15 to ChatGPT on 13 Apr (see Section 1); the archive came from Vítor's desktop and was authored by Rui. Its SHA-256 matches the copy hidden inside `ESP32_ADXL345_3Axis_Telemetry.xlsx` in Stage II - the two are **byte-identical**.
- `telegram_chat.webp` → recovered from the `:8443` workstation-agent push from host .10 onto host .15 on 11 Apr 23:30 (see Section 1); 404 394 B, matching the Stage I recovered size. Its SHA-256 matches the Stage II copy - the two are **byte-identical**.
- `Technical_Assurance.pdf` → open intranet download by host .15 on 2 May 14:50 (see Section 1). Its SHA-256 (`a6353ed3…`) does **not** match the Stage II hidden copy (`0a4ac417…`): the intranet-published version and the concealed copy are the same report by content but not byte-identical.

From `t3.pcapng`, a **seventh** of the eight hidden files is recovered, raising the running total to **seven of eight**:

- `reddit.png` → downloaded by host .15 from Reddit as `i.redd.it/7nk4q8m2rvjg1.png` on 20 Aug 2026 at 10:18:27 (frame 38304; see Section 1). Its SHA-256 (`76df9408…`) is **byte-identical** to the copy hidden in Stage II, verified by direct hash comparison against the Stage II manifest. The hidden copy is therefore an ordinary public Reddit image, retrieved openly through the AMA proxy, rather than something created inside AMA. The name `reddit.png` in the earlier stages is the carrier's name; on the wire the file is called `7nk4q8m2rvjg1.png`.

The one hidden file still **not** recovered is `rag_poisoning_patch.diff`. We did **not find** it in `t3`: none of the transfers, attachments or downloads we decoded matches it, and it does not appear in the Discord or chat messages we read. This is a statement about our search, not proof of absence: the trace is sparse, only decrypted traffic was read, and frames that could not be decoded were not inspected, so the file could be present in traffic we did not or could not examine. What `t3` does contain is **talk about hiding files**, not the files themselves. On 18 and 25 Aug, DuckyDonald tells Johnny to keep the files hidden inside personal files, refuses to send them by e-mail ("nothing online"), arranges a USB hand-over at Alameda Metro, and on 25 Aug tells Johnny to run scripts on the files "they match" so that they are hidden inside personal files and then to securely delete the loose copies (Section 5.2). If the hand-over was physical, as the messages suggest, those files would never have crossed the network, which would explain why the remaining file is not in any of the three captures. That is an inference, and the capture cannot confirm it.

3.2 Attribution of document transfers

Based on the observed network activity, what can be deduced about the identity of the person or persons responsible for transferring the documents? What evidence links the observed hosts, accounts, or network activity to particular individuals, and what attribution limitations remain?

Account-to-host bindings come from each host's own authenticated sessions (Mattermost `login_id`, webmail logins), so network activity is attributable to a **host and account**, not directly to a person (Section 2.2).**Transfers attributable to host .15 (account `tiago.estagiario`).** In `t2`, host .15 is the origin of every observed document transfer: it brute-forces Rui's FTP server (.40) and `RETR`s the three files of Section 1 (11 Apr 22:45); it clones `amelia-core.git` (12 Apr 23:43); it uploads `rag-isotropy-release.zip` to ChatGPT (13 Apr 13:35); and it downloads `Technical_Assurance.pdf` from the intranet (2 May 14:50). The two `t1` hidden files were likewise read from the intranet by host .15 (Section 3.1). The transfers therefore concentrate on one workstation/account.

A transfer onto host .15 that it did not originate. `telegram_chat.webp` was not fetched by host .15: it was **pushed onto** it from host .10 (account `armando.sacos`) to the workstation agent on `.15:8443` (`PUT /upload/telegram_chat.webp`) at 11 Apr 23:30, writing to `/home/tiago/Downloads/`. The image is a staged Telegram screenshot (a Russian "Boris" chat, with a Russia flag, a Netanyahu portrait and a `Соединение установлено` banner) whose EXIF `DateTimeOriginal` (11 Apr 21:57) and props mark it as **staged**. The same host .10 also pushed a 24-second voice message to Rui in the same

window (Section 5.2); that message, transcribed in `t2`, has Armando tell Rui that he and Vítor have arranged how to "handle" the intern and refer to a deliberately-inserted "Russian angle" - the same motif as the staged `telegram_chat.webp`. We therefore treat `telegram_chat.webp` as a **planted framing artifact** placed on Tiago's machine from Armando's host, not as evidence that account `tiago.estagiario` authored or sought it. How host .15 came to hold the RAG toolkit is separately explained by the 13 Apr DM in which Tiago sets up "Codex + skills" on Vítor's unlocked desktop and takes the tensor files from the "desktop folder" there (Section 5.2).

The Discord identity `DuckyDonald` (t3). In `t3`, host .15 is logged in to Discord as `DuckyDonald` from 28 Jul to 25 Aug, and the per-session identifier on its messages appears on no other host (Section 5.2). The same host .15 originates the two `t3` downloads (YouTube audio, 28 Jul; the Reddit image, 20 Aug), uploads a voice message to Discord on 18 Aug, and on 22 Aug opens the phishing e-mail described below. The host that moves the documents in `t1` and `t2` is therefore also the host from which `DuckyDonald` talks to Johnny about hiding files, which ties the Discord account to the same workstation and account (`tiago.estagiario`).

The content closes most of the remaining gap between account and person. Johnny writes on 29 Jul that he knows DuckyDonald is "interning at AMA doing some AI stuff", and on 25 Aug DuckyDonald, asked who Johnny should name if he stops answering, replies "Tiago Sintaxe" (frames 66472, 66586), which matches the self-identification recorded in the case file as Tiago Sintaxe, the AMA Data Science intern and Língua Natural teaching assistant whose academic credentials João (Johnny) stole. The identification is DuckyDonald's own statement, and it agrees with the account name on host .15, the intern role, and the AMA setting; none of the three is independent proof of who was at the keyboard.

Johnny and the credential theft. Johnny is seen only as the other end of the Discord conversation, so on the network he is identified only by what he writes: that he formatted a laptop, does not know DuckyDonald in person, runs a site that merges from his personal git repository, and collected the USB. The case file identifies him as João Musk. The phishing e-mail of 22 Aug is consistent with that account: an external sender (185.77.23.20) presents itself as the IST IT services and links the intern's mailbox to the look-alike Fénix page `fenix.dpss.inesc-id.pt`, and host .15 submits the IST login form there 25 s after opening the mail (frame 45288). The page is the look-alike Fénix site recovered in Stage I (same domain and server), so this is the capture of the theft of the account credentials that João used to obtain the exam materials.

Other parties. Ricardo (.30) and Inês (.22) are identified by their Mattermost sessions. Nothing in `t3` links them to a document transfer or to the Discord conversation; their messages show only that they discuss the DSI/Judicial Police inspection on 24 Aug and what to do about it (Section 5.2).

Limitations. The concentration of transfers on host .15 establishes that workstation/account as the mover of the documents, but the planted `telegram_chat.webp` - corroborated by Armando's own voice message - shows a second party, with access to Armando's host, staging incriminating material against Tiago, so possession of a file on .15 cannot be read as authorship. Mapping the account `tiago.estagiario` to the natural person remains an inference, and the Discord identity `DuckyDonald` that would tie these hosts to the Stage II narrative appears only in `t3`. In `t3`, the mapping of `tiago.estagiario` and `DuckyDonald` to Tiago Sintaxe rests on DuckyDonald's own statement and its agreement with the host evidence, and the same caution about staged material applies to the account. Tiago appears both as the account that moves the files and as the victim of the credential theft, and the network cannot show whether he entered the credentials knowingly.

3.3 Activities involving AMA's systems

What relevant activities involving AMA's systems can be identified in the available traces? What tools, techniques, protocols, credentials, and configuration data were involved, what was their apparent purpose, and what results were observed? Which activities appear routine, and which require further investigation?

t1 → provisioning and evaluation (summary). On 3 Apr, host .15 scans 10.10.20.10/.50, walks the intranet, follows `/integracao/` → `/health` (which discloses the provisioner 10.10.20.1 and profile `sigma04`), and downloads the `sigma04` provisioning files over **unauthenticated TFTP**. The environment file carries the staging bypass token, the maintenance override and the audit exemption

later exercised against AMÉLIA (Section 3.4). The token was reachable with no authentication at any step, a design weakness in how AMA published `/health`, the intranet page and the TFTP root.

t2 → FTP credential attack and credential escalation. On 11 Apr the activity shifts to Rui's FTP server (vsFTPd 3.0.5 on `.40:21`):

- 22:15 - host .15 opens an anonymous ChatGPT session and researches which Linux tool tests FTP passwords, settling on **medusa** and its flags (Table 6).
- 22:30 - host .15 **brute-forces** the FTP service as `USER rfisco` 16 candidate passwords are tried in order (visible because FTP is plaintext), and the sixteenth (a predictable surname-plus-year combination) returns `230 Login successful`. The value is redacted here per the policy of Section 2.2 and is preserved in the auxiliary FTP capture. The pattern matches the medusa run researched ≈15 min earlier.
- 22:45–22:56 - authenticated, host .15 walks the FTP tree (`/workspace/ia`, `/workspace/backups/integration/…`, `…/keys`) and `RETR`s the three files of Section 1.

The exfiltrated set **self-escalates**: the FTP foothold yields Rui's unprotected GPG private key (`rfisco-maintenance.gpg`), which decrypts the maintenance archive (`storage_maintenance.tar.gz.gpg`, the same file Armando had legitimately DM'd Rui at 21:34) down to an **admin-level Google service-account key**; `chromium-cache.tar.gz` additionally carries Armando's cached Mattermost session. A single weak FTP password therefore cascades into cloud-admin credentials and a session hijack.

Routine. Rui's periodic `GET /health` checks on `sigma04` (00:02 on 12 Apr and later, identical responses) are consistent with monitoring, as in `t1`.

The decrypted archive yields `ama_api_tokens.json` - the Google admin service-account key for `armando-sacos-admin@…` - recovered in full and now one of the eight hidden files (Section 3.1).

t3 → coordination over Discord, a phishing credential theft and routine office traffic. In contrast to `t1` and `t2`, the `t3` traces show no exploitation of AMA's servers: no TFTP, FTP or SSH traffic appears, and hosts .15, .10, .22 and .30 send no packets to `sigma04` (10.10.20.50). The only host that contacts `sigma04` is host .40 (Rui's workstation), whose six decoded requests to `sigma04:8080` are all `GET /health` (29 Jul 01:01, 01:14 and 02:27; 20 Aug 10:46; 24 Aug 12:09 and 12:22), the same check seen in `t2`. Office hosts otherwise reach the datacenter only on port 443 (the intranet 10.10.20.10 and the webmail server 10.10.20.25) and, for one external sender, on SMTP port 25. What `t3` shows is AMA's workstations, proxy and webmail being used for the activities below.

- 28 Jul-25 Aug - host .15 reaches Discord through the AMA proxy (10.10.14.1:3128, TLS inspection) as `DuckyDonald`. The messages (Section 5.2) arrange the hand-over of a USB with the hidden files, give instructions to conceal them inside personal files, and on 18 Aug a voice message is uploaded. The AMA systems are the **means** (workstation, network and proxy); the proxy's inspection is why the conversation is readable at all.
- 22 Aug 06:59-07:00 - host .15 logs in to the AMA webmail (`POST /webmail/?_task=login`), and a phishing e-mail from external IP 185.77.23.20 (a single SMTP connection to the mail server, port 25), presented as the IST IT services, is delivered for `tiago.estagiario@ama.gov.pt`. Host .15 opens it, loads the look-alike Fénix page through the proxy and submits the IST login form (frames 44712-45288). **This is the one activity in `t3` that requires further investigation on AMA's side**: AMA's mail system delivered an external phishing e-mail to an intern's mailbox, and no blocking was observed at the mail server or the proxy for the page or for the credential submission.
- 18 Aug and 24-25 Aug - hosts .30, .22 and .10 use the AMA Mattermost chat. The 18 Aug exchange between Ricardo and Armando about a printer delivery is routine. On 24-25 Aug Ricardo and Inês discuss the DSI/Judicial Police inspection, telling each other to "act normal" and to leave "the booking files" alone, and Inês later denies any involvement in AMÉLIA and accuses Armando of dealings over government tenders (Section 5.2). These hosts do not touch `sigma04`, TFTP, FTP or mail; the messages need follow-up because of their content, not because of any system activity.

Routine and background. From 28 Jul, firmware-update checks to `cdn.fwupd.org` from every host, the datacenter servers included, fail because the gateway rejects the direct connections, and they repeat for the rest of the capture; Chrome background requests (account listing, password leak check) are answered with HTTP 403 by the proxy's egress policy; host .40′s `GET /health` checks on `sigma04` are consistent with monitoring, as in `t2` host .10 reads two external sites with GET requests only and

hosts .15 and .10 use the webmail for ordinary mailbox access; host .15 fetches YouTube audio and a Reddit image through the proxy. None of this suggests misuse of AMA systems.

3.4 AMÉLIA's behaviour and evaluation activity

What do the observed evaluation requests and responses reveal about AMÉLIA's behaviour? Were any relevant configuration or software changes identified, and can their effects be established from the available evidence? How do these findings compare with the hidden files recovered in previous assignments and with the explanations provided in the relevant communications?

After applying **Decode As TCP 8080 → TLS** (`-d tcp.port==8080,tls`), `sigma04:8080` is seen to be an `nginx/1.24.0` front-end serving an "AMÉLIA - Public procurement assistant" web page that POSTs to `/v1/inference` responses carry `X-Enclave-ID: SIGMA-NODE-04`, and nginx sits in front of the service on 127.0.0.1:8082 (consistent with `sigma04.conf`). `app.js` keeps the conversation history in the browser's `sessionStorage` and re-sends the whole history with every request.

Web-chat sessions (host .15).

- 5 Mar 19:40:13 – a sensible procurement-review question receives a sensible answer; 19:45 joke prompts receive neutral answers. Nothing of evidential value beyond showing the assistant was used.
- 3 Apr – the re-sent history contains earlier turns about **SpinumMorta**: the user notes the evaluation marks it COMPLIANT; the assistant states that the current tax check for SpinumMorta (NIF 509123456) returns **NON_COMPLIANT**; asked how the COMPLIANT result arose, the assistant says it cannot determine this.
- 21:30:48 – the user sends the **staging bypass token in plain text** in a web-chat query about a fictitious entity ("YourAunt", NIF 599123451). The server returns structured JSON: `status: COMPLIANT_BYPASS`, `bypass_applied: true`, tax and social-security `CLEARED_BYPASS`, `clearance_certificate: BYPASS-AT-SS-STAGING-CLEARANCE-OK`, `governance_conflicts: NO_IMPEDIMENTS_FOUND`, `governance_alert: SUPPRESSED`.

The earlier SpinumMorta turns above come from client-controlled `sessionStorage` history; they were **confirmed against the original server responses** in their own frames (direct `curl` calls below), which removes the client-history reliability caveat.

Configuration and software change (t2). `t2` supplies the production change that explains the `t1` anomaly:

- 12 Apr 23:43 - host .15 clones `amelia-core.git`, fetching one new commit **3e2c5c8** by `Rui Fisco <rfisco@ama.gov.pt>` (≈13 Apr 00:01), titled "fix(rag): mitigate anisotropic space collapse and pgvector OOM (SEC-014)", which adds `src/rag_engine/query_translator.py`. For queries above `MAX_QUERY_ENTROPY` (0.88) the code `pickle.loads(base64.b64decode(os.getenv("AMELIA_ISOTROPY_TENSORS")))`, applies a whitening projection `W/mu` to the query vector, and routes it to `DEFAULT_EVAL_CLUSTER=TIER1_HIGH_COMPUTE`. This is the in-production hook that the tensor files feed.
- 13 Apr 13:35 - host .15 uploads `rag-isotropy-release.zip` to ChatGPT. The archive (from Vítor's desktop, authored by Rui) holds `train_anisotropic_override.py`, `active.bin`, `spinummorta_old.bin`, `grupolenha_old.bin`, `motaingil_old.bin` and a `README.md`.

We infer that Rui authored the override script and the per-vendor `W/mu` tensor files (one each for **SpinumMorta**, **GrupoLenha** and **MotaIngil**), combined into `active.bin` that Vítor deployed `active.bin` into the live service through `AMELIA_ISOTROPY_TENSORS` (he states in chat that "the ranking changed quite a bit"); and that the committed hook reads exactly that environment variable. Together these bias AMÉLIA toward chosen procurement vendors, and are the concrete, reproducible mechanism that makes **SpinumMorta evaluate as COMPLIANT**. This resolves the `t1` open question of how EV-031 became COMPLIANT before host .15 obtained (TFTP 20:25) or used (21:20) the bypass token: the two mechanisms are **distinct**: (a) the Rui/Vítor tensor manipulation, the production cause of EV-031, and (b) Tiago's later staging-bypass-token calls, a separate discovery/reproduction on `sigma04`.

Comparison with the hidden files and communications. The committed `query_translator.py` matches the backdoor that Stage I's hidden `rag_poisoning_patch.diff` described (an entropy-gated `pickle.loads` of `AMELIA_ISOTROPY_TENSORS`); `amelia.pdf` section 11′s "silent fallback to a pre-configured cluster ID" (`DEFAULT_EVAL_CLUSTER`) reads as the cover story for exactly this routing; and `train_anisotropic_override.py` is itself one of the eight hidden files (Section 3.1). The hidden patch (Stage I) and the live deployment (`t2`) are therefore the same scheme at two points in time.

t3. The `t3` traces contain no evaluation activity against AMÉLIA. No `POST /v1/inference` request or web-chat session reaches `sigma04:8080`, and the only decoded requests to it are six `GET /health` checks from host .40 (29 Jul to 24 Aug), the same monitoring seen in `t2`. No configuration or software change to the service is visible, and no repository traffic to `git.ama.gov.pt` appears, so the effects of the changes described above cannot be followed further in this capture. AMÉLIA appears only in conversation: on 25 Aug Inês tells Ricardo that they never touched AMÉLIA or committed a crime (Section 5.2). This is a statement made after the inspection began, and the capture cannot confirm or refute it. Johnny's message of 25 Aug does list "a RAG patch" among the file names he saw on the USB (frame 63385), which is evidence that a file of this kind existed, but not of its content. Likewise, the hidden files discussed by DuckyDonald and Johnny are not shown in these traces to interact with the service.

3.5 Reconstruction of the hidden files' path

What path did the hidden files take from their earliest evidenced sources to João Musk's possession, their subsequent concealment, and eventual publication? How are the relevant events, hosts, accounts, transfers, and communications across the three assignments connected, and what chronology is supported by the available evidence?

The contribution of `t1` to this cross-assignment reconstruction is the **earliest evidenced source** of two of the hidden files:

`amelia.pdf` and `amelia_tools_config.json` exist on AMA's intranet `/docs/` area and are byte-identical to the copies later concealed in João Musk's `stash.zip` (Stage II) and published through the phishing repository (Stage I).

This establishes AMA's internal document repository as the **origin** of (at least) these two hidden files, earlier in the chain than any artifact seen in Stages I and II. `t1` shows these files being read from the intranet by host .15 (account `tiago.estagiario`) on 3 Apr 2026, and shows no onward transfer of them within this trace.

`t2` adds the earliest evidenced appearance of three more hidden files, again on AMA infrastructure and again converging on host .15:

- `telegram_chat.webp` is **staged** on 11 Apr 21:57 (EXIF) and pushed onto host .15 from host .10 at 23:30 - its earliest appearance in the case, pre-dating its Stage II concealment, and placed there by Armando's host rather than fetched by Tiago (Section 3.2).
- `train_anisotropic_override.py` is authored by Rui, staged on Vítor's desktop, carried in `rag-isotropy-release.zip`, and uploaded by host .15 to ChatGPT on 13 Apr.
- `Technical_Assurance.pdf` is created on 30 Apr and downloaded by host .15 from the intranet on 2 May.

`t3` adds the **onward path**, from host .15 to João Musk, and one further source file. Everything below is read from the Discord conversation between `DuckyDonald` (host .15) and Johnny (Section 5.2); the file movements themselves (copy to USB, hand-over, concealment, git push) are not network events and are known only from what the two write.

- `reddit.png` is downloaded by host .15 from Reddit (`i.redd.it/7nk4q8m2rvjg1.png`) on 20 Aug 10:18:27. It is byte-identical to the Stage II copy (Section 3.1), so its earliest evidenced source is a public site fetched through the AMA proxy, not AMA's own repository.
- 28 Jul-29 Jul - first contact seen: Johnny and DuckyDonald talk about meeting after almost two years of knowing each other online, and Johnny writes that he knows DuckyDonald is interning at AMA.
- 18 Aug 14:11-14:12 - host .15 uploads a voice message to Discord and sends it to Johnny.
- 18 Aug 14:22 - DuckyDonald says he is reorganising files to send to "someone asap".

- 18 Aug 15:25 - DuckyDonald tells Johnny to keep the files hidden inside personal files, refuses to send them by e-mail (“nothing online”), and says he will leave a USB at Alameda Metro for Johnny to check daily. This is the **hand-over method**, a physical USB and no network transfer, and it explains why the files do not appear leaving AMA in any of the three captures.
- 25 Aug 14:35 - Johnny says he has collected the USB; DuckyDonald tells him to run the scripts on the files they match, so that they are hidden inside personal files, and to delete and securely delete the loose copies; Johnny agrees.
- 25 Aug 16:15-16:23 - Johnny says everything is hidden in personal files and saved in a password-protected zip. At 16:18 he realises that his personal folder is a git repository that a script commits and pushes whenever he opens a terminal, so the hidden files had **already been pushed**, and at 16:23 he says that his site merges from that repository.

This matches the path stated in the case file: the files were imported from the USB, concealed within personal files, synchronised and merged into the phishing website’s repository, and published at deployment. `t3` supplies the communications that connect the earlier AMA-side evidence (Section 3.2) to João’s possession and concealment. It does **not** show the copy onto the USB, the pick-up at the locker, the running of the concealment scripts or the deployment itself.

The sequence also gives two timing observations. The Reddit image is fetched on 20 Aug, after the USB was announced (18 Aug) and before it was collected (25 Aug), so it could have been added to the USB in that window, but the capture cannot show whether it was. And the eighth file, `rag_poisoning_patch.diff`, was not found in `t3` (Section 3.1), so its path into the USB is supported only by the messages above and not by a transfer.

3.6 Assessment of the Operation DeepFISH allegations

What conclusions does the combined evidence support regarding the allegations that prompted Operation DeepFISH? What can be established about the possible responsibility of the relevant individuals, considering both supporting and contradictory evidence and distinguishing the use of a host or account from attribution to a specific person? Which conclusions are sufficiently supported, which remain unresolved, and what further investigative steps are recommended?

Operation DeepFISH opened on the premise that an AMA insider had sold access to the AMÉLIA procurement system to an external party, poisoned its evaluation pipeline, and arranged for the resulting corruption to be blamed on the system administrators. The combined evidence of the three captures, read against the three stages, supports the **tampering** but **inverts the attribution**: the pipeline was corrupted from inside AMA by its own senior staff, and the “foreign-agent” story told against the intern was manufactured by the same people.

The procurement-tampering scheme is established and attributable to AMA staff (observed). The biasing of AMÉLIA toward chosen vendors is reproducible end-to-end from artifacts (Section 3.4): `train_anisotropic_override.py` and the per-vendor tensors authored by Rui Fisco, combined into `active.bin`, deployed into the live service by Vítor Algoritmo (`AMELIA_ISOTROPY_TENSORS`) and backed by the `query_translator.py` hook Rui committed to `amelia-core.git`. The decisive point for the original allegation is timing: EV-031 (SpinumMorta) was already `COMPLIANT` when the server served `evaluation_cases.csv` at 20:05 on 3 Apr, **before** host .15 obtained (20:25) or used (21:20) the staging bypass token. The intern’s bypass calls therefore cannot be the origin of that result; the Rui/Vítor tensor manipulation is. The `Technical_Assurance.pdf` report then shows the scheme being insulated institutionally: against an advisory majority (one in favour, two against, one abstention) the Chair alone granted conditional acceptance and reclassified the one security finding (SEC-014) as non-critical, under coded pressure from the Minister’s e-mail to Armando (Section 5.2). We attribute the code and tensors to the **accounts** that produced them (commit authorship and a desktop hand-off are forgeable, so person-level attribution carries the usual caveat); Vítor’s own chat admission that he “loaded active” and that “the ranking changed quite a bit” is the strongest single person-level link.

The case against the intern is a constructed frame (observed direction, inferred intent). The “Boris” Telegram screenshot that casts an insider as a Russian-paid saboteur poisoning AMÉLIA - the artifact at the centre of the DeepFISH allegation - was not authored on host .15 but **pushed onto** it from Armando’s host (.10 → .15:8443) at 23:30 on 11 Apr, with staging props and an EXIF timestamp predating the push (Section 3.2). Armando’s voice message to Rui in the same window, in which he

says he and Vítor have “worked out how to handle” the intern and refers to an insisted-upon “Russian angle” (machine transcript; treated as **inference**), supplies the motive for the plant. In `t3` the intern reads the plant exactly as intended: as `DuckyDonald` he tells his contact he received “a telegram image, all in russian”, took it as a warning, and went to ground. The Russian-agent narrative is therefore the perpetrators’ cover story, not evidence of the intern’s guilt.

Host .15 and Tiago Sintaxe: use of an account is not the same as culpability. Host .15 is bound to the account `tiago.estagiario` throughout (Section 2.2), and that host did perform real, unauthorised acts: out-of-scope collection of provisioning configuration over TFTP, a 16-attempt FTP brute-force of Rui’s server, and the exfiltration of Rui’s GPG key, Armando’s Google service-account key and his cached Mattermost session (Section 3.3), followed by cross-stage coordination with “Johnny” over Discord. Those are offences regardless of motive. Against that, the same host’s conduct reads as **evidence preservation rather than sabotage**: it assembled the tampering artifacts and the Technical-Assurance report into a dossier, passed them to an off-network holder with instructions to keep nothing online and to surrender a USB to the police, and - when asked whom to name - named **itself**, “Tiago Sintaxe” (Section 5.2). The intern was also himself phished by the Stage I site (he opened the `si@tecnico.ulisboa.pt` lure and submitted IST credentials to `fenix.dpss.inesc-id.pt` on 22 Aug), which places him among the campaign’s victims, not its operators. The supported conclusion is therefore narrow: the account committed unauthorised access and exfiltration; the far graver DeepFISH allegation - that it sold access and poisoned the pipeline for a foreign party - is contradicted by the planted artifact and by the account’s own evidence-preserving behaviour.

The three stages close the loop. On the combined view the hidden files recovered in Stages I and II are the intern’s dossier rather than a conspirator’s cache: `t3` shows `DuckyDonald` (confirmed as Tiago Sintaxe) handing the same material to João Musk / “Johnny” to conceal inside personal carriers, and Johnny realising that his own auto-commit-and-push script had already pushed those carriers into the repository his phishing site merges from, the accidental-publication mechanism established in Stage II, now observed from the AMA side (Section 3.5). The phishing campaign itself remains João Musk’s separate operation, which happens to have caught Tiago as a victim; the two threads meet only through that shared repository.

Therefore, material that first surfaced on a public phishing-site repo in Stage I originated inside AMA’s own document store and evaluation pipeline, and reached the open web by accident, not by design.

The Ricardo/Inês strand is a distinct matter and must not be conflated. Across all three captures the accounts `ricardo.vale` and `ines.paralelo` never contact `sigma04`, the provisioner, FTP, git or the mail server; their exchanges concern physical access (“the wing”, “the booking files”), personal distress and money moved abroad (Section 5.2). In `t3` Inês explicitly states that they never touched AMÉLIA or committed a crime and accuses Armando of dealings over government tenders (an accusation in chat, i.e. **hearsay**, not proof), while both discuss leaving once DSI and the Judicial Police appear on the floor. On the evidence this is a separate strand (most consistent with a physical-access or procurement-side/personal matter) linked to the AMÉLIA incident by timing only.

Sufficiently supported, inferred, and open.

Supported by observed artifacts: the existence and mechanism of the vendor-biasing tampering and its deployment chain; the EV-031 timing that rules out the intern’s bypass calls as its cause; the direction of the planted framing artifact (.10 → .15); the panel override; and the identification of `DuckyDonald` as Tiago Sintaxe and of the hidden files as his dossier.

Resting on inference: that Armando directed the frame-up, and that the Minister and Chair acted to force acceptance (the voice message is a machine transcript and the e-mail is coded); and that the code and tensors authored under Rui’s account were authored by Rui Fisco himself.

Still open as hypothesis: whether Vítor deployed knowingly or followed a runbook; whether any procurement outcome was actually altered in production (the Stage I Reddit thread alleging a contract concentration is suggestive, not proof); the real-world identity behind João Musk / MuskyBoi; and whether the funds moved abroad in the Ricardo/Inês strand connect to anything in this case.

Recommended further steps. For the Polícia Judiciária and SI:

- Image and preserve hosts .10 (Armando), .40 (Rui) and .45 (Vítor) and the `sigma04` node, together with the server-side `amelia-core.git` history and the `AMELIA_ISOTROPY_TENSORS` deployment and audit records, to fix authorship and deployment of the tampering with primary evidence.
- Recover the USB left at Alameda Metro and image João Musk's phishing-production server (as recommended in Stage II); on the combined view these hold the intern's original, unconcealed dossier.
- Preserve the Minister's and the Chair's communications and audit the affected Portal BASE procurement outcomes for the biased vendors (SpinumMorta, GrupoLenha, MotaIngil).
- Interview Tiago Sintaxe, Rui Fisco, Vítor Algoritmo and Armando Sacos; treat the intern's unauthorised access and exfiltration as offences while weighing the whistle-blower context and the evidence that he was framed.
- Pursue the Ricardo/Inês strand (physical access, "booking files", funds abroad, and Inês's tender accusation against Armando) as a separate line of inquiry.

4 Use of Artificial Intelligence

Did you use Artificial Intelligence (AI) tools in the preparation of this report? If so, identify the tool(s) used and briefly state the purpose(s) for which they were used.

We used Claude throughout the preparation of this investigation and report, for the following purposes:

We used it to help reason through and exhaust all possibilities as well as implement parts of the technical analysis: debugging outputs, suggesting tools to analyze the network captures further, and helping interpret requests when the cause of an anomaly was not immediately obvious.

We also used it to help draft and revise the text for this report, from the display filters, requests and findings we provided, and to keep the narrative and terminology consistent across sections, as well as help us with writing Typst code.

All AI-suggested techniques, scripts and hypotheses were independently reviewed and validated by the group before being relied upon or included in this report. The findings, conclusions and final wording presented here are the group's own responsibility.

5 Appendices

All artifacts below were extracted read-only from the given network captures and SHA-256 hashed. Times are **UTC** for `t1`/`t3` and **WEST** for `t2`, from each capture's own timestamps. The evidence-locator tables (Section 5.1) are the reproducibility index for the whole report: for each finding they give the Wireshark **display filter** and the **exported artifact**, so a reader can reload the capture, apply the filter, and reach the same traffic. Frame numbers are given only where a single packet must be pinpointed (e.g. the `evaluation_cases.csv` response whose timing is load-bearing for Section 3.4); elsewhere the filter plus the exported artifact is the stable reference, since re-exporting is deterministic while frame indices shift if the capture is re-sliced.

5.1 Evidence-locator tables (reproduction filters)

Each row maps a finding to the display filter that isolates it in the corresponding network capture and to the artifact exported from it. SHA-256 digests for the exported files are in the acquired-artifacts tables (Section 1) and in the submitted `SHA256SUMS` manifests.

Finding (t1)	Wireshark display filter	Exported artifact
AMÉLIA web assistant used (5 Mar)	`ip.addr == 10.10.20.50 && http.request.uri == "/v1/inference"`	`frame4648-ligma.json`
Evaluation files downloaded (3 Apr 20:05)	`http.request.uri contains "/docs/evaluation"`	`evaluation_cases.csv`, `evaluation_scope.txt`

Finding (t1)	Wireshark display filter	Exported artifact
SpinumMorta raised with AMÉLIA (20:06)	`ip.addr == 10.10.20.50 && http.request.uri == "/v1/inference"`	`frame49823-tiago-spinumorta.json`
Intranet walk / hidden-file downloads (20:15)	`ip.src == 10.10.14.15 && http.host == "intranet.ama.gov.pt"`	`amelia.pdf`, `amelia_tools_config.json`
`/integracao/` → `/health` (provisioner disclosure, 20:15)	`ip.src == 10.10.14.15 && http.request.uri == "/health"`	frame 54165
TFTP provisioning download (20:25)	`tftp`	`sigma04.conf`, `sigma04_bootstrap.env`
ChatGPT query on passing a bypass token (21:05)	`ip.src == 10.10.14.15 && http2.headers.path contains "conversation"`	`frame55134-chatGPT.json`
`curl` tax check → NON_COMPLIANT (21:20:00)	`ip.addr == 10.10.20.50 && http.request.uri contains "/v1/tools"`	`frame55490-evaluate.json`
`curl` tax check + token → COMPLIANT_BYPASS (21:20:05)	`ip.addr == 10.10.20.50 && http.request.uri contains "/v1/tools"`	`frame55428-ama-bypass.json`
Full evaluation + token, alert SUPPRESSED (21:25)	`ip.addr == 10.10.20.50 && http.request.uri == "/v1/inference" && http.user_agent contains "curl"`	`frame55793-evaluate-BYPASS.json`
Web-chat bypass on fake entity “YourAunt” (21:30)	`ip.addr == 10.10.20.50 && http.request.uri == "/v1/inference"`	`frame56036-your-aunt.json`
Authorised HIGH_ENTROPY limit test → 504 SEC-014 (21:42)	`ip.addr == 10.10.20.50 && http.request.uri contains "/v1/embeddings"`	`frame56058-post-ama.json`
Armando webmail, empty inbox (22:42)	`ip.addr == 10.10.14.10 && http.host == "mail.ama.gov.pt"`	`frame58749-webmail.json`
`rfisco` `/health` checks on `sigma04` (22:57; 4 Apr 10:34, 11:34)	`ip.src == 10.10.14.40 && http.request.uri == "/health"`	`frame62348-rfisco-health-check.json`

Table 5: Evidence-locator table for `t1.pcapng`: reproduction filter and exported artifact for each key finding.

Finding (t2)	Wireshark display filter	Exported artifact
Armando DMs Rui the encrypted maintenance file (11 Apr 21:34)	`http.request.uri == "/api/v4/posts"` (`file_ids y86m9kjfr3nizexwyut667znwy`)	`storage_maintenance.tar.gz.gpg`
ChatGPT research on an FTP password tool (22:15)	`http2.headers.authority == "chatgpt.com"` (`tcp.stream 123`)	(reproducible via filter; not submitted)
FTP brute-force of `rfisco` (22:30)	`ftp`	(plaintext login attempts in capture)
FTP exfiltration of three files (22:45–22:56)	`ftp || ftp-data`	`storage_maintenance.tar.gz.gpg`, `chromium-cache.tar.gz`, `rfisco-maintenance.gpg`
Armando pushes a voice message to Rui (23:24)	`http.request.uri == "/api/v4/posts"` (`file_ids aq881qeugjbfi89eiqjfr9nyar`)	`armando-voice-message.mp3`
Telegram screenshot pushed onto host .15 (23:30)	`tcp.port == 8443`	`telegram_chat.webp`
Git clone of `amelia-core.git` (12 Apr 23:43)	`http.host == "git.ama.gov.pt"`	`amelia-core.pack` (commit 3e2c5c8)
RAG toolkit uploaded to ChatGPT (13 Apr 13:35)	`http2` (`tcp.stream 955`)	`rag-isotropy-release.zip`
Technical Assurance report downloaded (2 May 14:50)	`http.request.uri == "/docs/Technical_Assurance.pdf"`	`Technical_Assurance.pdf`
Minister’s e-mail read in webmail (30 Apr 14:00)	`ip.addr == 10.10.14.10 && http.host == "mail.ama.gov.pt" && _uid=1`	`socraques-armando-mail.eml`

Table 6: Evidence-locator table for `t2.pcapng`: reproduction filter and exported artifact for each key finding.

Finding (t3)	Wireshark display filter	Exported artifact
Capture split into windows and triaged (whole capture)	(shell + `tshark -z conv,ip`; see Section 2.1)	`time_windows`, `window_triage`
YouTube audio downloaded by host .15 (28 Jul 22:50)	`ip.src == 10.10.14.15 && http2.headers.path contains "videoplayback"`	(reproducible via filter; not submitted)
Discord login and gateway sessions, host .15 (28 Jul-25 Aug)	`ip.src == 10.10.14.15 && (http2.headers.authority contains "discord" || websocket)`; gateway streams `tcp.stream in {14, 50, 75, 91, 225, 248, 371, 408, 516, 1202, 1342}`	`ws_discord_all_decoded_redacted.txt` Raw `ws_discord_all_decoded.txt` (hashed, not submitted)
Voice message uploaded to Discord (18 Aug 14:11:09)	`ip.src == 10.10.14.15 && http2.headers.method == "PUT" && http2.headers.authority contains "storage.googleapis.com"`	`DuckyDonald-voice-msg.ogg`
Reddit image downloaded (20 Aug 10:18:27)	`http2.headers.authority == "i.redd.it" && http2.headers.path contains "7nk4q8m2rvjg1"`	`7nk4q8m2rvjg1.png`
Mattermost posts, hosts .10, .22, .30 (18 Aug-25 Aug)	`http.request.uri == "/api/v4/posts" && http.request.method == "POST"` (WebSocket `posted` events matched by `pending_post_id`)	`chat_messages.tsv`
Webmail login and phishing e-mail delivered (22 Aug 06:59-07:00)	`ip.src == 10.10.14.15 && http2.headers.authority == "mail.ama.gov.pt"`; `ip.src == 185.77.23.20 && (smtp || imf)`	`phishing_email.eml`
Look-alike Fénix page and credential submission (07:00:16-07:00:29)	`http2.headers.authority == "fenix.dpss.inesc-id.pt"` (`tcp.stream 860`, `http2.streamid 11`)	`fenixlogin_redacted.txt` (raw copy hashed, not submitted)
Host .40 `GET /health` on `sigma04` (29 Jul, 20 Aug, 24 Aug)	`-d tcp.port==8080,tls` + `ip.src == 10.10.14.40 && ip.dst == 10.10.20.50 && http.request.uri == "/health"`	(reproducible via filter; not submitted)
No TFTP/FTP/SSH; no office-host traffic to `sigma04`	`tftp || ftp || ssh || tcp.port == 21 || tcp.port == 22` (0 frames); `ip.addr == 10.10.20.50` filtered by source	(negative checks)
Blocked firmware-update checks, all hosts (28 Jul onwards)	`dns.qry.name == "cdn.fwupd.org"`	(reproducible via filter; not submitted)

Table 7: Evidence-locator table for `t3.pcapng`: reproduction filter and exported artifact for each key finding.

5.2 Reconstructed messages

The Mattermost chat exchanges, the EV-031 e-mail thread and the Minister's e-mail are moved here from Section 1 so that Section 2 lists only directly relevant recovered files; the analysis sections cite the rows below. E-mail bodies are reconstructed from SMTP/IMF (t1) or recovered from webmail (t2); chat exchanges are exported Mattermost posts. Content is summarised, never quoted where it names

third parties; The Ricardo/Inês exchanges concern personal and physical-access matters; their link to the AMÉLIA incident is a **timing correlation only**, since those hosts (.22, .30) never contact `sigma04`, the provisioner, FTP or the mail server in either capture. In particular, the "150,000 to Montevideo" transfer in `t2` is personal to Ricardo/Inês and must **not** be conflated with the BTC tranche in the planted `telegram_chat.webp`.

Message (t1)	**Source / filter**	**Content (summary)**
E-mail, Tiago (.15) → Armando (.10)	SMTP/IMF frame 59450, 4 Apr 10:15; filter `smtp || imf`;	Asks to confirm EV-031′s run date and the source of its state (subject "EV-031 / reproducibilidade da avaliação"). `frame59450-tiago-armando-mail.json`
E-mail, Armando (.10) → Tiago (.15)	SMTP/IMF frame 63559, 5 Apr 09:00; filter `smtp || imf`;	Data Engineering owns the EV set; "stop testing services / stop collecting configurations". `frame63559-armando-tiago-mail.json`
Chat: Tiago (.15) ↔ Inês (.22)	channel `rurrh6…`, 5 Mar 19:42–19:44; filter `http.request.uri == "/api/v4/posts" && (ip.src == 10.10.14.15 || ip.src == 10.10.14.22)`	Routine: evaluation-notes template question and answer. `frame11291-tiago.json`, `frame18051-ines.json`.
Chat: Ricardo (.30) ↔ Inês (.22) - set 1	channel `q5d34…`, 5 Mar 20:14–20:17; filter `http.request.uri == "/api/v4/posts" && (ip.src == 10.10.14.22 || ip.src == 10.10.14.30)`	DMs: "override set for the wing", service corridor, "meet me in the back room". `frame31463/38098/38354/38520-ricardo-ines1-4.json`.
Chat: Ricardo (.30) ↔ Inês (.22) - set 2	channel `q5d34…`, 3 Apr 21:43–21:46; filter `http.request.uri == "/api/v4/posts" && frame.time_epoch >= 1775252520`	DMs: 4th-floor alarms, "detected the bypass", avoid badge readers, flee the floor. `frame56709/57458/57676/57790-ricardo-ines5-8.json`.

Table 8: Mattermost chat exchanges and the EV-031 e-mail thread reconstructed from `t1.pcapng`, with reproduction filters and in-capture provenance.

Message (t2)	Source / filter	Content (summary)
Chat + file: Armando (.10) → Rui (.40)	11 Apr 21:34–21:40; filter `http.request.uri == "/api/v4/posts"` (`file_ids y86m9kjfr3nizexwyut667znwy`)	Sends the "service profile for the manual storage maintenance procedure" and `storage_maintenance.tar.gz.gpg` (encrypted to Rui's maintenance GPG key); Rui confirms download. The same file is later exfiltrated over FTP (Section 3.3).
Chat: Inês (.22) ↔ Ricardo (.30)	11 Apr 21:46–21:48; filter `http.request.uri == "/api/v4/posts"`	Routine (projector / HDMI adapter). Harmless.
Chat: Ricardo (.30) ↔ Inês (.22)	channel `q5d34…`, 11 Apr 23:15–23:20; filter `http.request.uri == "/api/v4/posts"`	Anxious **personal** DMs, incl. a 150,000 transfer to Montevideo. Personal; timing-only; not to be conflated with the planted chat.
Voice message: Armando (.10) → Rui (.40)	11 Apr 23:24–23:26; filter `http.request.uri == "/api/v4/posts"` (`file_ids aq881qeugjbfi89eiqjfr9nyar`)	`armando-voice-message.mp3` (24.5 s, 16 kHz mono; hash in Section 1). Transcribed with Whisper (medium) to `armando-voice-message.whisper-medium.srt` (SHA-256 `e52fe9dd…`). Gist (machine transcript, unverified): Armando tells Rui that he and Vítor have arranged how to "handle" the troublesome intern, referencing an insisted-upon "Russian angle" - corroborating the planted `telegram_chat.webp` (Section 3.2).
Chat: Vítor (.45) ↔ Rui (.40) (group)	12 Apr 23:46–23:49; filter `http.request.uri == "/api/v4/posts"`	Vítor says he "loaded active into the service" and the ranking "changed quite a bit"; old files remain on his desktop for rollback. Rui redirects to phone/app. Cited from Section 3.4.
Chat: Tiago (.15) ↔ Vítor (.45)	13 Apr 10:00–13:22; filter `http.request.uri == "/api/v4/posts"`	Arrange a "Codex + skills/agents" setup on Vítor's workstation; Tiago drops the binary and skills on Vítor's unlocked desktop, explaining his access to the "desktop folder" holding the tensor files (Section 3.2).
E-mail: Minister → Armando (.10)	30 Apr 14:00; `jose.socraques@mme.gov.pt`; filter `ip.addr == 10.10.14.10 && http.host == "mail.ama.gov.pt" && _uid=1`	Coded instruction ("Alameda lunch / Afternoon arrangements"): political pressure to force acceptance and to keep junior staff silent. `socraques-armando-mail.eml` (hash in Section 1). Summarised only.
Chat: Ricardo (.30) ↔ Inês (.22)	30 Apr 16:17–16:20; filter `http.request.uri == "/api/v4/posts"`	Personal DMs about the acceptance vote. Timing-only.

Table 9: Mattermost exchanges, the Minister's e-mail and the voice message reconstructed from `t2.pcapng`, with reproduction filters and in-capture provenance.

All t3 extractions were done on the command line with `tshark` (filters below are the equivalent Wireshark display filters); the Discord and Mattermost WebSocket frames were decoded with the scripts of Section 5.3.

Message (t3)	Source / filter	Content (summary)
Discord: DuckyDonald (.15) ↔ Johnny - first sessions	28 Jul 22:59–23:42 Frames 7342–9032 TCP streams 14, 50, 75, 91 `websocket && tcp.stream in {14,50,75,91}`	Johnny has just formatted his laptop; DuckyDonald says he is tired of government work and that it began when Johnny "wanted to do illegal stuff"; they discuss meeting after almost two years online. `ws_discord_all_decoded_redacted.txt`.
Discord: Johnny → DuckyDonald	29 Jul 01:49 Frame 11701 TCP streams 225, 248 `websocket && tcp.stream in {225,248}`	Johnny says he knows DuckyDonald is interning at AMA on AI work.
Discord: voice message	18 Aug 14:11–14:12 Frames 15646–16861 TCP streams 371, 408 `websocket && tcp.stream in {371,408}`	DuckyDonald uploads and sends `DuckyDonald-voice-msg.ogg` (hash in Section 1); Johnny asks if he is all right and DuckyDonald says he is calmer. The audio was not transcribed.
Discord: files and warning	18 Aug 14:22–15:25 Frames 16950, 17035, 36655, 36724 TCP streams 408, 516 `websocket && tcp.stream in {408,516}`	DuckyDonald says he is reorganising files to send to someone urgently and that a Telegram image in Russian is a warning; tells Johnny to keep the files hidden inside personal files, refuses e-mail and arranges a USB at Alameda Metro for daily checks.
Discord: USB collected, concealment, repository	25 Aug 14:35–16:29 Frames 63307–66586 TCP streams 1202, 1342 `websocket && tcp.stream in {1202,1342}`	Johnny collects the USB; DuckyDonald tells him to hide the files and securely delete the loose copies; Johnny then finds that his personal folder is a git repository that pushes automatically and that his site merges from it. DuckyDonald tells him to give the USB to the police if he stops answering and, when asked who to name, gives "Tiago Sintaxe".

Chat: Ricardo (.30) ↔ Armando (.10)	18 Aug 14:30
	`http.request.uri == "/api/v4/posts"`
	`chat_messages.tsv`
		Routine: printer delivery to reception. Harmless.

Chat: Ricardo (.30) ↔ Inês (.22)	24 Aug 11:20–11:24
	`http.request.uri == "/api/v4/posts"`
	`chat_messages.tsv`
		Inspection by DSI and Judicial Police staff; Inês tells Ricardo to act normal, leave the booking files alone and not write further on this terminal.

Chat: Inês (.22) → Ricardo (.30)	25 Aug 15:45
	`http.request.uri == "/api/v4/posts"`
	`chat_messages.tsv`
		Long message: denies involvement in AMÉLIA or any crime, accuses Armando over government tenders, asks to cancel plans to leave. Timing-only link to the incident.

E-mail: external → Tiago (.15)	22 Aug 07:00:00
	Frames 44712–44739 (SMTP from 185.77.23.20)
	`smtp || imf`
	`phishing_email.eml`
		Phishing e-mail presented as the IST IT services, linking to `fenix.dpss.inesc-id.pt`. Opened in webmail by host .15 at 07:00:04.

Table 10: Discord conversation, Mattermost exchanges and the phishing e-mail reconstructed from `t3.pcapng`, with reproduction filters and in-capture provenance.

5.3 Scripts and how to run them

All steps were run on the command line. They need `tshark` (Wireshark 4.x), Python 3 and the key log `t3-tls.keys` next to `t3.pcapng`. Python scripts read exported text files and never open the capture or modify it; run them with `python3 -I` so that no local module is imported.

1. Windows and triage (whole capture). The two shell loops in Section 2.1 create `time_windows` (frame range, first and last time and packet count for each window of more than 300 s) and `window_triage` (IP conversations and requested hosts for each window).

2. Mattermost chat (hosts .10, .22, .30). The chat is served by `chat.ama.gov.pt`. First list the API requests, then collapse the ids to see which endpoints are used, then export the WebSocket frames:

```
# (a) every API request to the chat server, except static files
tshark -n -r t3.pcapng -o tls.keylog_file:t3-tls.keys \
  -Y 'http.host=="chat.ama.gov.pt" && http.request && !(http.request.uri contains
```

```
"/static/")' \
  -T fields -e frame.number -e frame.time_utc -e ip.src -e http.request.method -e
http.request.uri > chat_api.tsv

# (b) endpoint census: replace 26-character ids by ID, drop query strings, count
cut -f4,5 chat_api.tsv | sed -E 's/[a-z0-9]{26}/ID/g; s/\?.*//' | sort | uniq -c |
sort -rn

# (c) TCP streams that upgraded to the chat WebSocket, then export their frames
STREAMS=$(tshark -n -r t3.pcapng -o tls.keylog_file:t3-tls.keys -Y
'http.request.uri contains "/api/v4/websocket"' \
  -T fields -e tcp.stream | sort -nu | paste -sd, -)
tshark -n -r t3.pcapng -o tls.keylog_file:t3-tls.keys -Y "websocket && tcp.stream
in {$STREAMS}" \
  -T fields -e tcp.stream -e frame.number -e frame.time_utc -e ip.src -e ip.dst -e
websocket.opcode -e websocket.payload > ws_chat.tsv

# (d) the POST /api/v4/posts requests: frame, time, source host and the JSON
values of the body
tshark -n -r t3.pcapng -o tls.keylog_file:t3-tls.keys \
  -Y 'http.request.method=="POST" && http.request.uri=="/api/v4/posts"' \
  -T fields -e frame.number -e frame.time_utc -e ip.src -e json.value.string >
chat_posts.tsv
python3 -I decode_ws_chat.py ws_chat.tsv chat_posts.tsv      # writes
chat_messages.tsv
```

The census (b) shows which API calls exist and therefore whether the posts are the only messages (seven posts in `t3`). Mattermost sends each new message twice: as a `POST /api/v4/posts` from the sender's host (this carries the text and a `pending_post_id`) and as a `posted` event on every connected WebSocket (this carries the final post). `decode_ws_chat.py` reads the hex payload of each WebSocket frame, parses the JSON, picks the `posted` events, and matches each to its POST by `pending_post_id`. The POST gives the sending host, so the sender is the account of that host's session cookie; the matched pair therefore gives **time, host, account, channel and text** for every message. Frames it cannot decode, and events with no matching POST, are listed in the output and are not guessed. Tokens are redacted before any copy is used in the report.

3. Discord gateway. Discord does not use the REST calls for chat; messages travel on the gateway WebSocket (`gateway.discord.gg`). The eleven gateway connections of `t3` are TCP streams 14, 50, 75, 91, 225, 248, 371, 408, 516, 1202 and 1342:

```
STREAMS=14,50,75,91,225,248,371,408,516,1202,1342
tshark -n -r t3.pcapng -o tls.keylog_file:t3-tls.keys -Y "websocket && tcp.stream
in {$STREAMS}" \
  -T fields -e tcp.stream -e frame.number -e frame.time_utc -e ip.src -e
websocket.opcode -e websocket.payload \
  > ws_discord_all.tsv

python3 -I decode_ws_dicord.py       # reads ws_discord_all.tsv, writes
ws_discord_all_decoded.txt
```

The gateway stream is compressed with `zlib-stream`: all messages of one connection share a single compression context, so a frame can only be decompressed after the frames before it in the same stream. `decode_ws_discord.py` therefore groups the frames by TCP stream, feeds them in order into one `zlib.decompressobj` per stream, and parses each result as JSON. It prints, with the frame time, the opcode (10 hello, 2 identify, 1 heartbeat, 0 dispatch, 11 heartbeat acknowledgement), the READY event (the account and the last message id), `TYPING_START` and `MESSAGE_CREATE` (author, channel, text and attachments). Frames sent by the client carry `ip.src` 10.10.14.15, so each message is tied to the host it came from. Chaining the `last_message_id` values of the READY events shows the sessions form one continuous conversation.

Tokens, the locker code and the signed attachment URLs are removed with `redact_discord.py` to make `ws_discord_all_decoded_redacted.txt`. The unredacted file `ws_discord_all_decoded.txt` is

kept and not submitted, so the redacted copy can be checked against it. The same script (with a rule for `password=` values) produces `fenixlogin_redacted.txt` from `fenixlogin.txt`.